Model Broadcast in TLA$^+$

DARE Project Presentations
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Background

Background

Approach

Background → Approach → Broadcast


Background
Approach
Broadcast
Implementation

Background → Approach → Broadcast → Implementation → Wrap up

Temporal Logic of Actions: TLA$^+$

- Specification language for distributed and asynchronous systems
  - Introduced by Lamport, based on temporal logic
- Built on first-order logic and set theory
- Captures system behavior via:
  - $\mathtt{Init}$ and $\Box[\mathtt{Next}]_{\langle \mathtt{vars} \rangle}$
  - Actions describing state transitions
  - Safety and liveness properties
- Operators for model checking
  - $\Diamond F$, $\Box G$, $\Diamond\langle A \rangle_v$, $F \leadsto G$, $\Diamond\Box F$, $\Box\Diamond F$

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  - Extended control flow: `if`, `elsif`
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- Language constructs:
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  - Extended control flow: `if`, `elsif`
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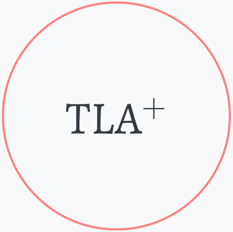
- Labeling rules ensure correct translation:
  - Mandatory in `while`-loops and after `either`, `call`, `return`
  - Required between consecutive writes to the same variable

Landscape



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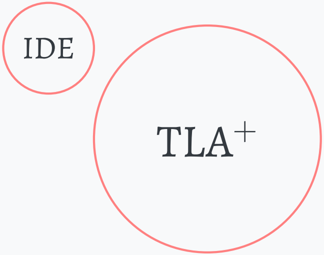

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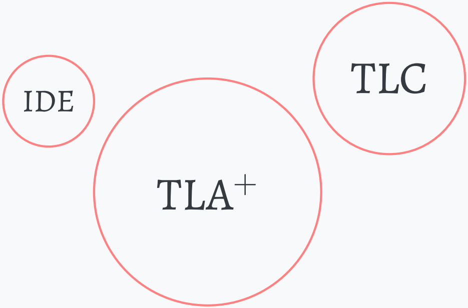

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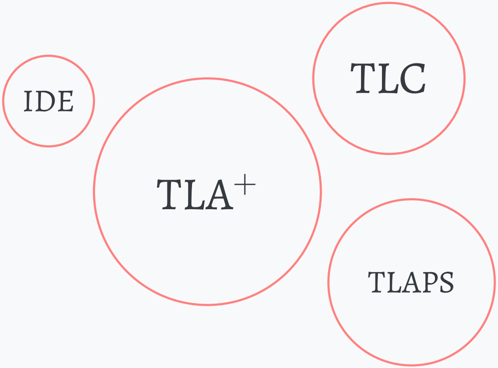

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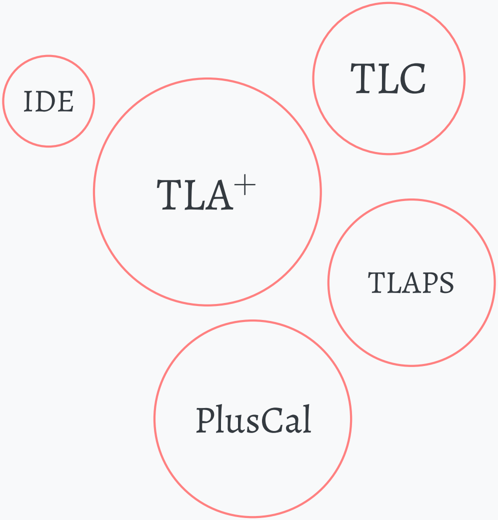


Approach

PlusCal algorithm

Approach

PlusCal algorithm

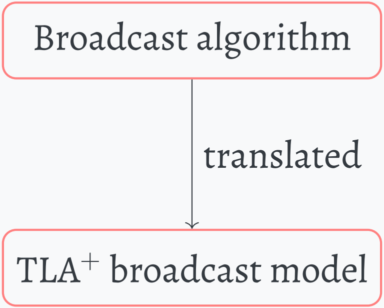
$\downarrow$ translated

TLA$^+$ spec

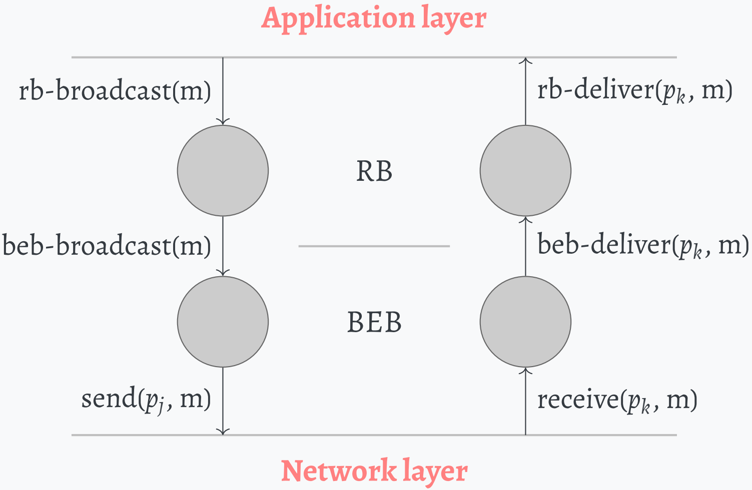
Approach

Broadcast algorithm

Approach



Broadcast recap



Broadcast can be done:

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- Best effort

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- Reliable

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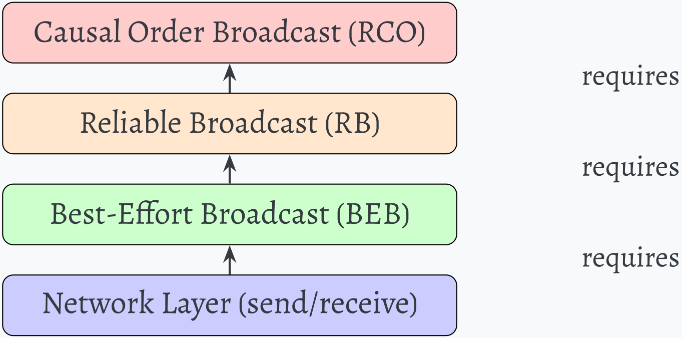
- Causal

Implementation Approach

Layered architecture: Each broadcast module builds upon the previous layer



Process Architecture

Two-process design: Separation of concerns between broadcasting processes and messaging (actor)

Broadcast Process(es)

- Implements broadcast algs
- Handles protocol logic

delegate send

Actor Process

- Sends messages
- Cannot broadcast

The broadcast process uses the actor process for actual network communication

Code demo for broadcast algorithm

...

A simple property

$$P_{EventualDelivery} = \Diamond \left(\bigwedge_{n \in Nodes} |delivered[n]| \neq 0 \right)$$

where

$Nodes = \{P_1, P_2, P_3\}$ and

$delivered = \{m_1, m_2, \ldots\}$

Conclusions

Pros

- Easy to use
- Fast to develop

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- Easy to use
- Fast to develop

Cons

- Less expressive compared to TLA$^+$
- It is not like Erlang
  - message sending setup
- Harder defining properties
  - due to TLA$^+$ model

Thank you for your ***ATTENTION***